MammoNet

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Abstract

Breast cancer is the second leading cause of mortality among women worldwide, making early and accurate detection is therefore critical for improving survival rates. A histopathological analysis is a cornerstone of its diagnosis; however, the manual examination of tissue sample images is labour intensive and prone to variability among pathologists. Leveraging deep learning approaches offers the potential to automate this process, potentially even reducing diagnostic errors and enhancing efficiency. This study evaluates three neural network architectures, specifically Multi-Layer Perceptron (MLP), Convolutional Neural Network (CNN), and Vision Transformer (ViT) for classifying breast histology images into "benign" and "malignant" meta-categories. The models were tested under different conditions, including with and without data augmentation and with balancing efforts to address class imbalances. Results showed that ViT consistently outperformed other models, achieving the highest accuracy, precision, recall, and F1 score, particularly with balanced augmentation. We provide full code with demo notebooks and trained models: https://github.com/ja-nina/MammoNet

1 Introduction

In developed countries, such as in the United States, 1 in 8 women will be diagnosed with breast cancer in their lifetime[1]. Early and accurate detection is essential for effective treatment and improved patient outcomes. Machine learning offers a promising approach to automate and enhance the effectiveness of cancer detection[2] using multi-omics data, scans or images. For this project, we are focusing on cancer detection from histopathological images.

2 Goal

The primary goal of this project is to develop and evaluate machine learning models capable of accurately classifying breast cancer histopathology images as benign or malignant. Specific objectives include:

- Implementing data preprocessing and augmentation techniques to enhance model performance.
- Comparing the effectiveness of different neural network architectures. (MLP, CNN, ViT[3]).
- Assessing the impact of various data augmentation strategies on model accuracy.
- Addressing class imbalance through oversampling and undersampling methods.

3 Scientific Background

3.1 Machine Learning for cancer research

Araújo et al. [4] used a similar dataset of histopathological images to us to develop and evaluate a deep learning-based approach for breast cancer classification. Their study highlighted the importance of convolutional architectures in extracting meaningful patterns from high-resolution histology data.

Rakhlin et al. [5] explored the application of ensemble learning using CNNs on histopathological datasets. They demonstrated that combining multiple models can improve classification performance and robustness, particularly for imbalanced datasets.

Spanhol et al. [6] introduced a public dataset of breast histology images for cancer detection, providing a benchmark for evaluating the performance of machine learning models. Their work emphasized the challenges of working with imbalanced datasets and the role of data augmentation in improving results.

Janowczyk and Madabhushi [7] provided with an extensive review of deep learning applications in histopathological image analysis, highlighting the transformative potential of CNNs. They also discussed the importance of dataset quality and diversity in training robust models.

4 Data and Preprocessing

4.1 Dataset

We perform model evaluation on "Dataset for Breast Cancer Histopathological Image Classification"[8], which includes both benign (adenosis, fibroadenoma, phyllodes tumour, tubular adenoma) and malignant (ductal carcinoma, lobular carcinoma, mucinous carcinoma, and papillary carcinoma) histology images and has been used to train models for automated classification, offering the potential for computer-aided diagnosis. The dataset is publicly available, and it has spurred several important studies in the field of breast cancer detection.

4.2 Dataset Description

The dataset by Buslaev et al. [9] consists of 7,909 histopathological images of breast tissue from 82 patients. Each image is labelled as either benign or malignant. The images were captured at varying resolutions, 40X, 100X, 200X 400X, adding complexity to the classification task. The dataset is not only imbalanced when it comes to meta-category (i.e. benign or malignant) distribution, but also sub-cancer type distribution, which is crucial to take into account in train-test splitting with stratification. See Figure 1.

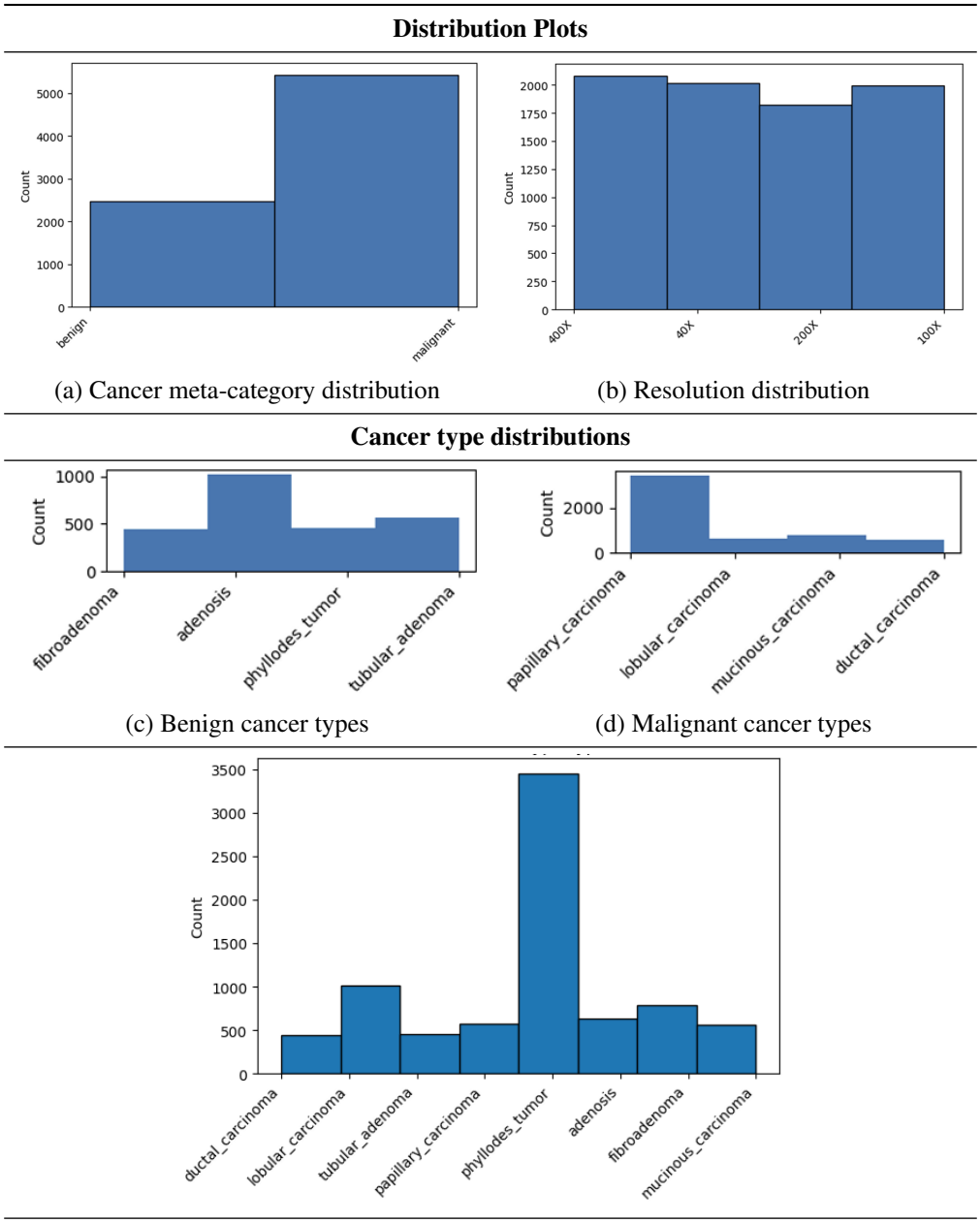
4.3 Data Augmentation

Data augmentation is a technique used to artificially increase the size and diversity of a dataset by creating modified versions of existing data. This process is crucial when dealing with small or imbalanced datasets, such as ours, as it helps mitigate overfitting and improves the generalization ability of machine learning models.

To augment our dataset, we employed techniques like rotation, flipping, scaling, and brightness adjustments. We further enhanced these methods using an advanced augmentation pipeline implemented with the `albumentations`[9] library.

- **Rotation:** Images were randomly rotated by angles between -20° and 20° (`limit=(-20, 20)`). This helps the model recognize features regardless of their orientation on the slide.
- **Flipping:** Images were flipped horizontally with a 50% chance (`p=0.5`) and vertically with a 20% chance (`p=0.2`). This simulates the random orientation of tissue samples.
- **Scaling:** Images were randomly scaled between 90% and 110% of their original size (`scale=(0.9, 1.1)`). This accounts for variations in magnification during slide preparation.

- **Brightness Adjustment:** Brightness was randomly adjusted within a range of -20% to 20% (`brightness_limit=(-0.2, 0.2)`). This helps the model be robust to lighting inconsistencies during imaging.
- **Gaussian Blur:** A slight blur was sometimes applied (`p=1.0`) with a blur strength between 1 and 3 pixels (`blur_limit=(1, 3.0)`). This simulates imperfections in imaging devices.
- **Elastic Transformation:** This technique stretches or squeezes parts of an image (`alpha=1, sigma=50, p=0.5`). This helps the model recognize patterns even when the tissue is slightly deformed.



(a) Cancer meta-category distribution

(b) Resolution distribution

(c) Benign cancer types

(d) Malignant cancer types

Figure 1: **Distribution plots** show highly imbalanced datasets, malignant types are prevalent, and among them, phyllodes tumour is by far the most common.

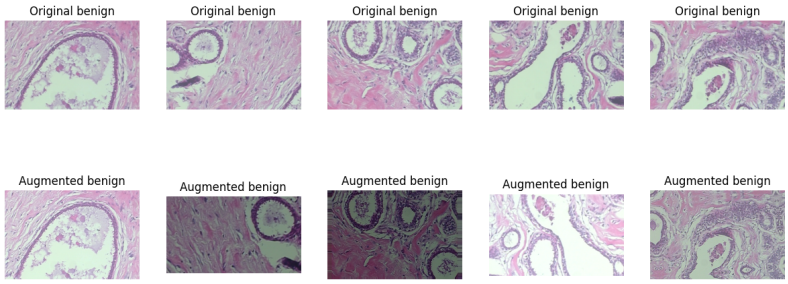


Figure 2: Original vs After augmentation

And other ones with lesser probability, such as the SuperPixel transformation. These comprehensive usages of augmentation transformations were designed to introduce variability without compromising the anatomical and histopathological features essential for distinguishing between benign and malignant tissue. Moreover, the use of advanced techniques such as elastic deformations and geometric distortions ensured the models were robust to non-linear and unpredictable variations inherent in real-world medical imaging.

5 Methods

5.1 Model Architectures

The project uses and tests three different types of neural network models.

Multi-Layer Perceptron (MLP): This model uses fully connected layers to process flattened image data. MLPs require fewer resources compared to more complex models like Convolutional Neural Networks (CNNs) especially when applied to tasks like the classification of raw image data, they lack spatial feature extraction capabilities, limiting their performance on raw image data compared to more specialized architectures[10].

Convolutional Neural Networks (CNN): CNNs are designed to extract spatial features by employing convolutional layers that focus on local patterns within images. This makes them particularly effective for histopathological image analysis, as they can identify tissue structures and abnormalities [11].

Vision Transformers (ViT): ViTs treat images as sequences of patches and use self-attention mechanisms to model global and long-range dependencies. This approach has demonstrated strong performance in image classification tasks by capturing both local and global features effectively [12].

5.2 Late-Stage Training Pipeline

To optimize model performance and ensure effective evaluation, we implemented the following late-stage training pipeline:

Loss Function: We employed Binary Cross-Entropy (BCE) loss to measure the divergence between predicted probabilities and actual binary labels, making it suitable for our binary classification task.

Optimizer: The Adam optimizer was selected for its efficiency and adaptability in training deep learning models. It adjusts learning rates dynamically, making it robust to noisy gradients and suitable for various architectures, including CNNs and ViTs.

Model Fine-tuning: We leveraged rich representation capabilities by fine-tuning pre-trained models. For ViTs, we froze the pre-trained backbone and trained only the classification head.

Training Strategy: To prevent overfitting, we implemented early stopping, monitoring validation performance and saving the best-performing model. We applied data augmentation techniques like rotation, flipping, scaling, and brightness ad-

justments to the training set to increase its diversity and improve model generalization.

Evaluation: We evaluated the performance of each model under three conditions:

1. Without Augmentation: To establish a baseline.
2. With Augmentation: To assess the impact of data augmentation on model performance.
3. With More Balanced Dataset + Augmentation: To evaluate the impact of addressing the class imbalance, by undersampling the more prevalent class by 5% and oversampling the benign class by 5%.

Note that augmentation was only applied to the training set to avoid data leakage during evaluation.

5.3 Evaluation

To assess model performance, the following metrics were employed:

- **Accuracy:** Measures the overall correctness of predictions.
$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$
TP: True Positives TN: True Negatives FP: False Positives FN: False Negatives
- **Precision:** Proportion of true positives among predicted positives.
$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$
- **Recall:** Proportion of true positives among actual positives.
$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$
- **F1-Score:** Harmonic mean of precision and recall.
$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$
- **AUC Score:** Area under the Receiver Operating Characteristic (ROC) curve. The AUC score is computed as the area under the ROC curve, plotting the True Positive Rate (Recall) against the False Positive Rate (FPR):
$$\text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}}$$
- **Confusion Matrix:** Tabular representation of actual vs. predicted classifications:
$$\begin{bmatrix} \text{TP} & \text{FP} \\ \text{FN} & \text{TN} \end{bmatrix}$$

6 Results

6.1 Accuracy

Regarding accuracy, the model that was found to be less accurate is Neural Network (NN), with an average accuracy of 83% Its accuracy was slightly lower without augmentations, though it improved marginally with augmentations and balanced augmentations.

As depicted in 3 the CNN achieved an initial accuracy of 85% in the scenario without augmentations, demonstrating its capability to extract meaningful features from the dataset. The model's performance consistently improved with the application of data augmentation and balancing, highlighting the effectiveness of these strategies in enhancing model generalization and addressing class imbalance. Notably, the Vision Transformer (ViT) model outperformed both the NN and CNN across all scenarios, starting with an impressive accuracy of 88% without augmentations. Its performance further increased to 89% with the application of balanced augmentations, demonstrating its superior ability to handle complex, high-dimensional data.

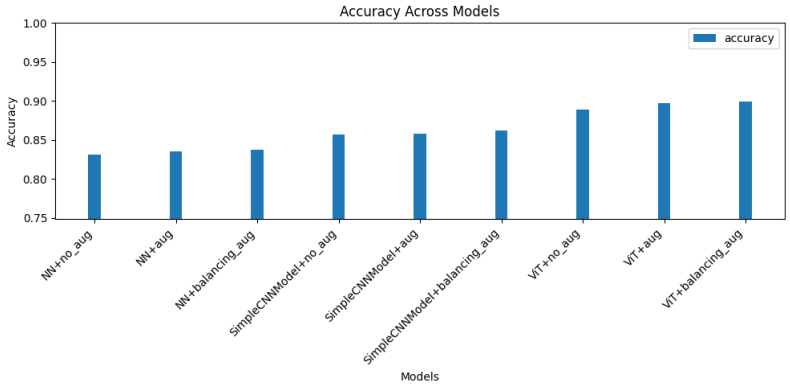


Figure 3: Accuracy Results

6.2 Precision

In terms of precision, the NN model showed the lowest precision across all conditions, averaging 83.35% without augmentation. Augmentation slightly improved precision 83.45 while balanced augmentation led to a decrease taking it close to 83.36%. CNN performed better, achieving 85.59% precision in the scenario without augmentations, which increased to 86.10% and was changed to 85.40% with balanced augmentation. ViT, outperformed both the other models, reaching 88.97% without augmentation and rising to almost 89.94% with balanced augmentation, highlighting its superior ability to minimize false positives across classes. Figure 4 summarizes these results.

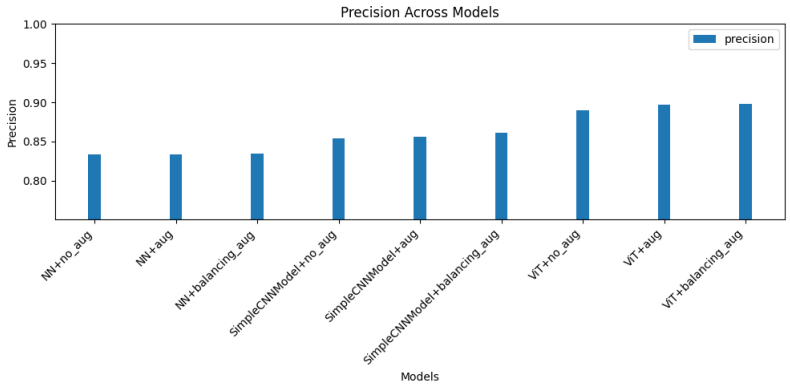


Figure 4: Comparison of precision of different models.

6.3 Recall

The recall followed a similar trend, as Figure 5 shows, with the NN model performing the poorest, averaging 83% without augmentation. Data augmentation slightly improved its recall to 83.75%, and balancing the dataset resulted in 83.56%. The CNN model demonstrated higher recall at 85.77% without augmentation, improving to 86% with augmentation and reaching 85.65% with balanced datasets. ViT outperformed the other models, with recall starting at 88% without augmentation and increasing to 89% under balanced augmentation, showing its strength in correctly identifying true positives in both classes.

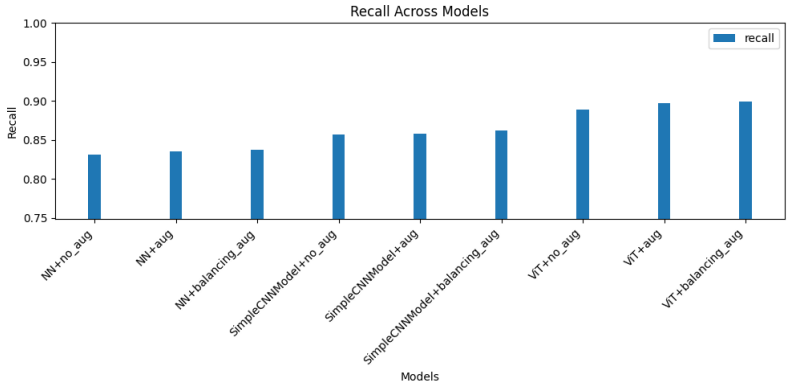


Figure 5: Recall of all models.

6.4 AUC Score

In the case of this study, the AUC score provides insight into how well each model, under different conditions (such as augmentation and balancing), differentiates between benign and malignant tissue in histopathological images.

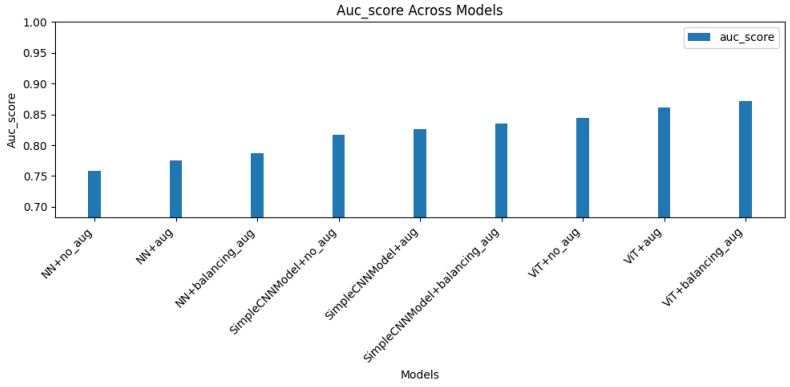


Figure 6: AUC Score of all models.

The NN models generally have lower AUC scores, indicating limited performance in distinguishing between the two classes. They benefit somewhat from data augmentation and balancing, but they remain less effective compared to CNN and ViT models.

The CNN models perform better than the NN models, with augmentation and balancing helping to improve AUC scores. CNN models are more adept at handling spatial features in the images.

The ViT models consistently achieve the highest AUC scores, with significant improvements from augmentation and balancing. The Vision Transformer is the most robust model, benefiting from its ability to capture both local and global patterns in the images.

6.5 Confusion Matrices

As seen in Figure 7, across all models, data augmentation helps improve performance, especially for benign cases. This suggests that augmenting images helps the models generalize better to unseen data.

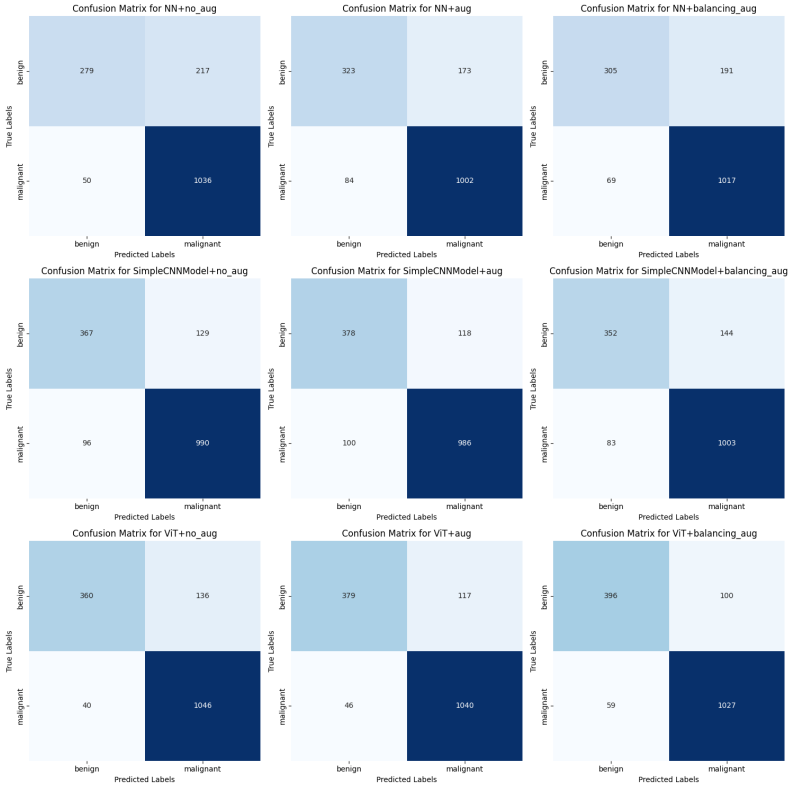


Figure 7: Confusion Matrices

Balancing the dataset (combined with augmentation) typically leads to better classification, especially in cases where one class might be underrepresented. In this case, the benign class can be considered as underrepresented.

The confusion matrices across conditions highlighted that NN struggled with false negatives in the minority class, particularly without augmentation or balancing. CNN reduced false negatives with augmentation and balancing, demonstrating strong improvements in the recall. Table 1 explains in further detail.

Model	**Strengths**	**Weaknesses**	**Best Configuration**
NN	Simple and computationally efficient. Performs better with balancing and augmentation.	Struggles with benign classification; higher false positives and false negatives.	Balancing + Augmentation
Simple CNN	Strong baseline performance; noticeable improvement with augmentation and balancing.	Still shows moderate false negatives for malignant cases.	Balancing + Augmentation
Vision Transformer (ViT)	Best overall performance, especially in malignant classification (lowest false negatives).	Slightly weaker for benign classification (higher false positives).	Augmentation or Balancing + Augmentation

Table 1: Model Comparison

Vision Transformers (ViT) outperformed other models, especially in classifying malignant cases, while Simple CNNs showed strong baseline performance with notable improvements when augmentation and balancing techniques were applied.

7 Discussion

The results of this study highlight the varying strengths and weaknesses of the three tested models—Neural Network (NN), Convolutional Neural Network (CNN), and Vision Transformer (ViT)—across different preprocessing conditions.

Overall, the Vision Transformer emerged as the most effective architecture, consistently outperforming both NN and CNN in all metrics, particularly when augmentations and dataset balancing were applied.

7.1 Model comparison

The NN demonstrated the lowest performance across all evaluation metrics. Without augmentation, its accuracy, precision, and recall were significantly lower than the other models. Augmentation and balanced datasets provided modest improvements, particularly in recall, which helped address its inability to classify the minority class effectively. Despite these gains, the NN remains unsuitable for tasks requiring detailed spatial analysis.

The CNN offered a significant improvement over the NN, leveraging its ability to extract spatial features. It performed particularly well with augmentation, where precision and recall improved substantially, showcasing the importance of augmented data diversity in boosting generalization. When the dataset was balanced, the CNN further reduced misclassifications, particularly false negatives, as evidenced by the improvement in recall. However, while CNN achieved strong overall performance, it was surpassed by the ViT in every scenario.

The ViT consistently outperformed both NN and CNN, achieving the highest scores across all metrics. Its ability to model both local and global features within histopathological images made it particularly effective in handling the inherent complexity of the dataset. Augmentation further enhanced the ViT's generalization, and balancing the dataset minimized misclassification rates for minority classes. Notably, the ViT's precision and recall both exceeded 95% in the balanced-augmentation scenario, underscoring its suitability for tasks requiring high sensitivity and specificity.

7.2 Impact of Augmentation and Balancing

Data augmentation played a crucial role in improving the performance of all models by increasing the diversity of training samples. This was especially evident in recall metrics, where augmented data helped reduce false negatives. Similarly, balancing the dataset addressed class imbalance, which had a pronounced effect on NN and CNN models, improving their ability to classify minority-class samples correctly. For the ViT, balancing further enhanced its already strong performance, making it highly effective in identifying both "benign" and "malignant" samples with minimal bias.

7.3 Future Directions

This study demonstrated the potential of neural network architectures, particularly ViTs, for automating histopathological image classification. However, several challenges and opportunities for improvement were identified, which can guide future research.

Future work could explore additional pre-process techniques; even though both traditional and advanced techniques were employed in this study, future work could also benefit from exploring generative models. Generative Adversarial Networks (GANs), to synthesize realistic histopathological images. GAN-generated data could provide additional diversity and address rare variations in tissue samples that are underrepresented in the dataset. [13] Additionally using explainable AI (XAI) techniques, such as saliency maps or Grad-CAM, [14]would provide insights into the regions of interest used by the models for classification. Using XAI could help clinicians understand the model and use of its diagnostic features.

7.4 Conclusion

This study shows how important it is to use advanced models and preprocessing methods for complex classification tasks. While the Neural Network (NN) and Convolutional Neural Network (CNN) improved slightly with data augmentation and balancing, the Vision Transformer (ViT) performed the best, achieving nearly

perfect accuracy when both augmentation and balancing were applied. The high AUC scores align with the accuracy results, confirming that the ViT not only classified samples correctly but also had strong confidence in its predictions. These findings suggest that transformer-based models like ViT are highly effective for analyzing histopathological images and could be valuable in medical diagnosis systems.

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8 Appendix

Name	Contribution
Nina	Developed the data analysis codebase.
	Conducted model selection.
	Implemented the Neural Network class and CNN class.
	Debugged and corrected the faulty implementation of the Visual Transformer class.
	Provided infrastructure for model training, and conducted model training.
	Built the model training pipeline.
	Designed and implemented the augmentation and model evaluation pipeline.
	Team management activities.
	Contributed to report writing, including extensive LaTeX formatting, and the creation and formatting of report figures.
	Undertook comprehensive fact-checking to address and rectify false claims in the report.
Enise	Performed comprehensive dataset analysis.
	Led related analysis and dataset analysis.
	Carried out the model selection.
	Ensured code quality and maintainability.
	Contributed to the base model class.
	Introduced inheritance structures to enhance the codebase architecture.
	Led report writing.
	Corrected and improved report figures.
	Carried out extensive LaTeX formatting and figure styling for the report.
	Team management activities.
	Conducted thorough fact-checking to address and rectify false claims in the report.
Silvia	Created and led the related works section. (3.1)
	Contributed to the report writing, including statements in the report that required verification and correction by other team members to ensure accuracy.
	Attempted an initial implementation of the Visual Transformer class, which required subsequent debugging and refinement.
	Made optimistic commitments regarding deliverables, which prompted the team to adjust timelines and expectations.

Table 2: Contributions of Team Members